

Lab04_Lambertucci

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Q1.

```
library(readxl)
library(tidyverse)
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v dplyr      1.2.1      v readr      2.2.0
v forcats    1.0.1      v stringr    1.6.0
v ggplot2    4.0.3      v tibble     3.3.1
v lubridate  1.9.5      v tidyr      1.3.2
v purrr      1.2.2
-- Conflicts ----- tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

Q2.

```
fname <- "WDIEXCEL.xlsx"
```

Q3.

```
countries <- read_excel(fname, sheet = "Country")
```

Q4.

```
countries <- countries %>%
  filter(`Short Name` %in% c("Brazil", "Argentina", "Chile",
                             "United States", "Germany"))
```

Q5.

```
WDI <- read_excel(fname, sheet = "Data")
```

Q6.

```
WDI <- WDI %>%  
  select(`Country Code`, `Indicator Code`, `1960`:`2018`)
```

Q7.

Não os dados não estão no formato Tidy, cada ano é uma coluna não uma linha

```
WDI <- WDI %>%  
  pivot_longer(  
    cols = `1960`:`2018`,  
    names_to = "YEAR",  
    values_to = "value"  
  )
```

Q8.

```
class(WDI$YEAR)
```

```
[1] "character"
```

```
WDI <- WDI %>%  
  mutate(YEAR = as.integer(YEAR))  
  
class(WDI$YEAR)
```

```
[1] "integer"
```

Q9.

```
WDI_wide <- WDI %>%  
  pivot_wider(  
    names_from = `Indicator Code`,  
    values_from = value  
  )
```

```
WDI_final <- WDI_wide %>%
  inner_join(countries %>% select(`Country Code`, `Short Name`),
            by = "Country Code") %>%
  select(`Country Code`, `Short Name`, YEAR, SP.DYN.CBRT.IN)
```

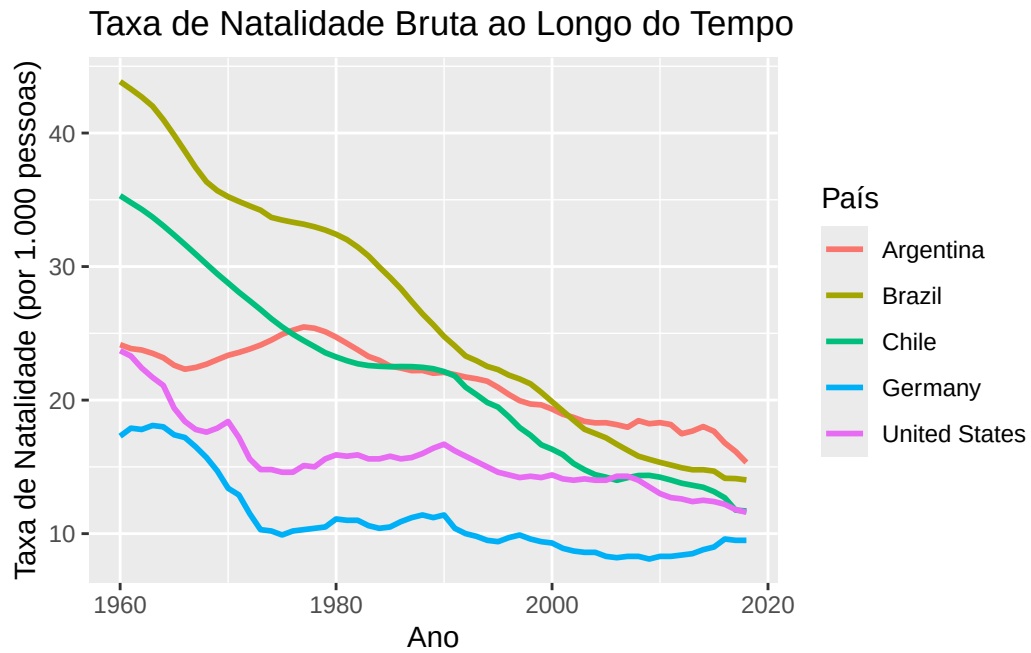
Q10.

```
WDI_final <- WDI_final %>%
  arrange(`Country Code`, YEAR)
WDI_final
```

```
# A tibble: 295 x 4
  `Country Code` `Short Name`  YEAR SP.DYN.CBRT.IN
  <chr>          <chr>        <int>      <dbl>
1 ARG           Argentina    1960      24.2
2 ARG           Argentina    1961      23.9
3 ARG           Argentina    1962      23.8
4 ARG           Argentina    1963      23.5
5 ARG           Argentina    1964      23.2
6 ARG           Argentina    1965      22.6
7 ARG           Argentina    1966      22.3
8 ARG           Argentina    1967      22.4
9 ARG           Argentina    1968      22.7
10 ARG          Argentina    1969      23.0
# i 285 more rows
```

Q11.

```
ggplot(WDI_final, aes(x = YEAR, y = SP.DYN.CBRT.IN,
                     color = `Short Name`)) +
  geom_line(linewidth = 1) +
  labs(
    title = "Taxa de Natalidade Bruta ao Longo do Tempo",
    x = "Ano",
    y = "Taxa de Natalidade (por 1.000 pessoas)",
    color = "País"
  )
```



Hora e Data

```
Sys.setenv(TZ = "America/Sao_Paulo")  
Sys.time()
```

```
[1] "2026-09-07 20:50:06 -03"
```